Three adaptive tactile modes

Automatically selected by real-time EEG state classification

Alerting mode

Trigger: sustained theta elevation + beta suppression (mind wandering)

IRON POWDER BEHAVIOUR

Electromagnets fire a sudden pulse (not gradual). Iron powder snaps into sharp clusters beneath the membrane, creating abrupt raised ridges under the palm. The sudden texture change is the key — gradual shifts are ignored by the somatosensory system.

NEUROSCIENCE MECHANISM

Abrupt tactile stimuli trigger phasic alerting — a transient spike in arousal mediated by the locus coeruleus-norepinephrine system. Light, sharp touch activates different nerve fibres than deep pressure, producing alertness rather than calm (Posner & Petersen, 1990).

Evidence: Tactile stimulation produces significant increases in EEG activation and heart rate (Laureys et al., 2007). Phasic alerting via cross-modal tactile cues confirmed in ADHD populations (Mayer et al., 2021, PMC8153018).

Soothing mode

Trigger: elevated beta + high EMG + gamma burst (anxiety / hyperarousal)

IRON POWDER BEHAVIOUR

Electromagnets cycle in a slow sine wave (3-5 second period) sweeping sequentially from index-finger side to pinky side. Iron powder forms a gentle moving wave across the palm surface. Simultaneous moderate baseline magnetic force provides sustained deep pressure.

NEUROSCIENCE MECHANISM

Slow stroking at 1-10 cm/s activates C-tactile (CT) afferent fibres, triggering oxytocin release and cortisol reduction. Deep pressure stimulation (DPS) activates the parasympathetic nervous system, shifting from fight-or-flight to rest-and-digest.

Evidence: CT-optimal touch reduces cortisol significantly ($p<0.0001$) in stressed populations (ScienceDirect, 2024). Weighted blankets improve sleep and emotional regulation in ADHD children (Occupational Therapy International, 2023). DPS increases dopamine and serotonin while reducing cortisol (Champagne et al., 2015).

Fidget / play mode

Trigger: TBR near threshold, baseline cognitive state (maintenance)

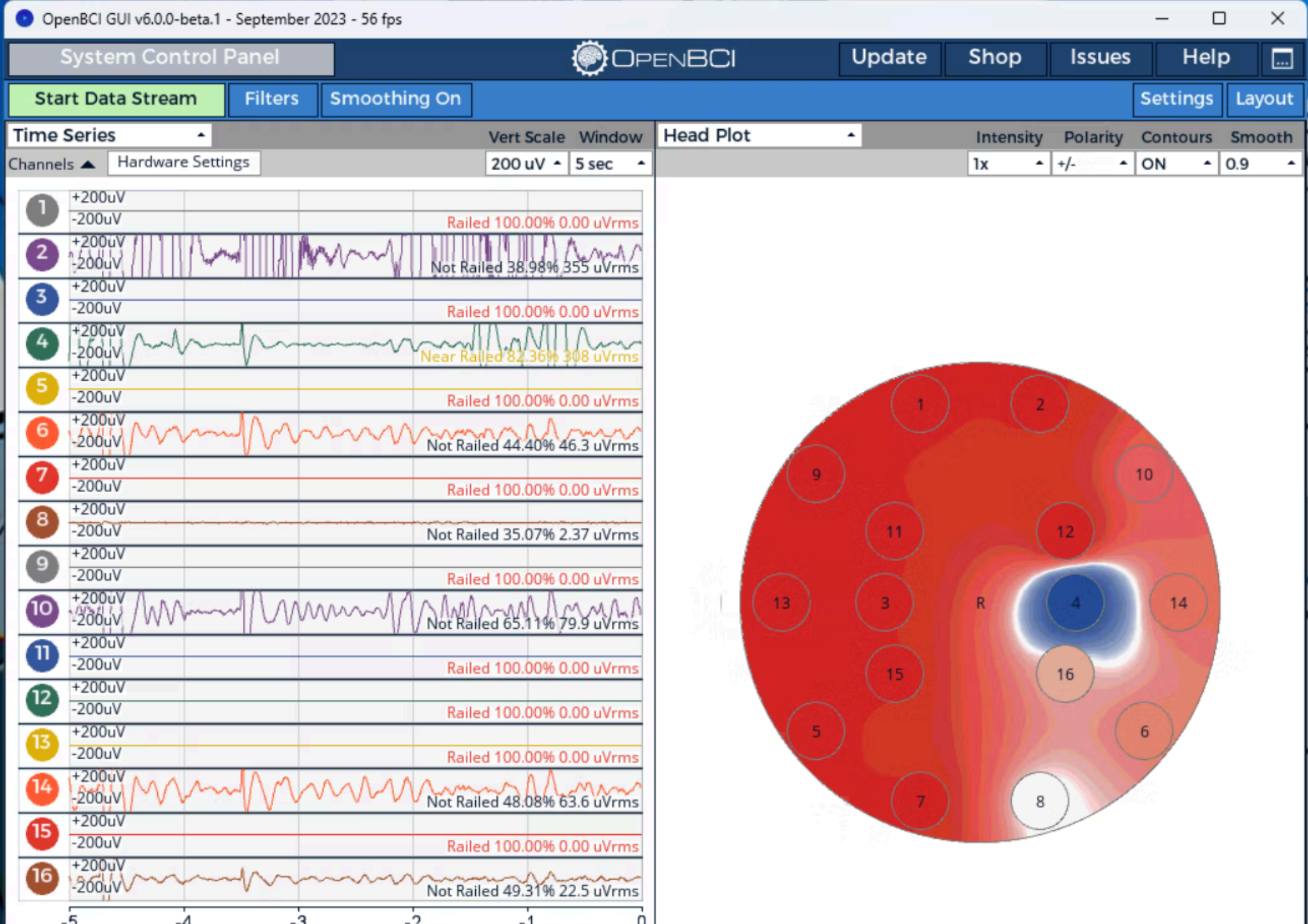
IRON POWDER BEHAVIOUR

Electromagnets hold moderate constant force — iron powder has a "baseline tension." User can squeeze, knead, and press the membrane freely. Pressed areas scatter; released areas re-gather. Electromagnet intensities slowly drift randomly, so each squeeze feels slightly different.

NEUROSCIENCE MECHANISM

ADHD involves chronic hypoarousal (Optimal Stimulation Theory, Zentall 1983). Movement compensates by raising arousal to the cognitive optimum. Bidirectional interaction (user pushes, material responds) resists habituation — unlike static fidget tools whose effect vanishes after first use.

Evidence: Higher motor activity predicts better working memory in ADHD but not in TD children (Sarver et al., 2015). Fidget spinners lose effectiveness after first exposure (Graziano et al., 2020). ADHD patients retain weighted/tactile tools longer than other groups (Swedish registry study, 2021).


OpenBCI GUI v6.0.0-beta.1 - September 2023 - 56 fps
System Control Panel
OpenBCI
Update
Shop
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Start Data Stream
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Settings
Layout
Time Series
Vert Scale
Window
Channels
Hardware Settings
200 uV
5 sec
Head Plot
Intensity
Polarity
Contours
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1x
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ON
0.9
+200uV
-200uV
Railed 100.00% 0.00 uVrms
Not Railed 38.98% 355 uVrms
Railed 100.00% 0.00 uVrms
Near Railed 82.36% 308 uVrms
Railed 100.00% 0.00 uVrms
Not Railed 44.40% 46.3 uVrms
Railed 100.00% 0.00 uVrms
Not Railed 35.07% 2.37 uVrms
Railed 100.00% 0.00 uVrms
Not Railed 65.11% 79.9 uVrms
Railed 100.00% 0.00 uVrms
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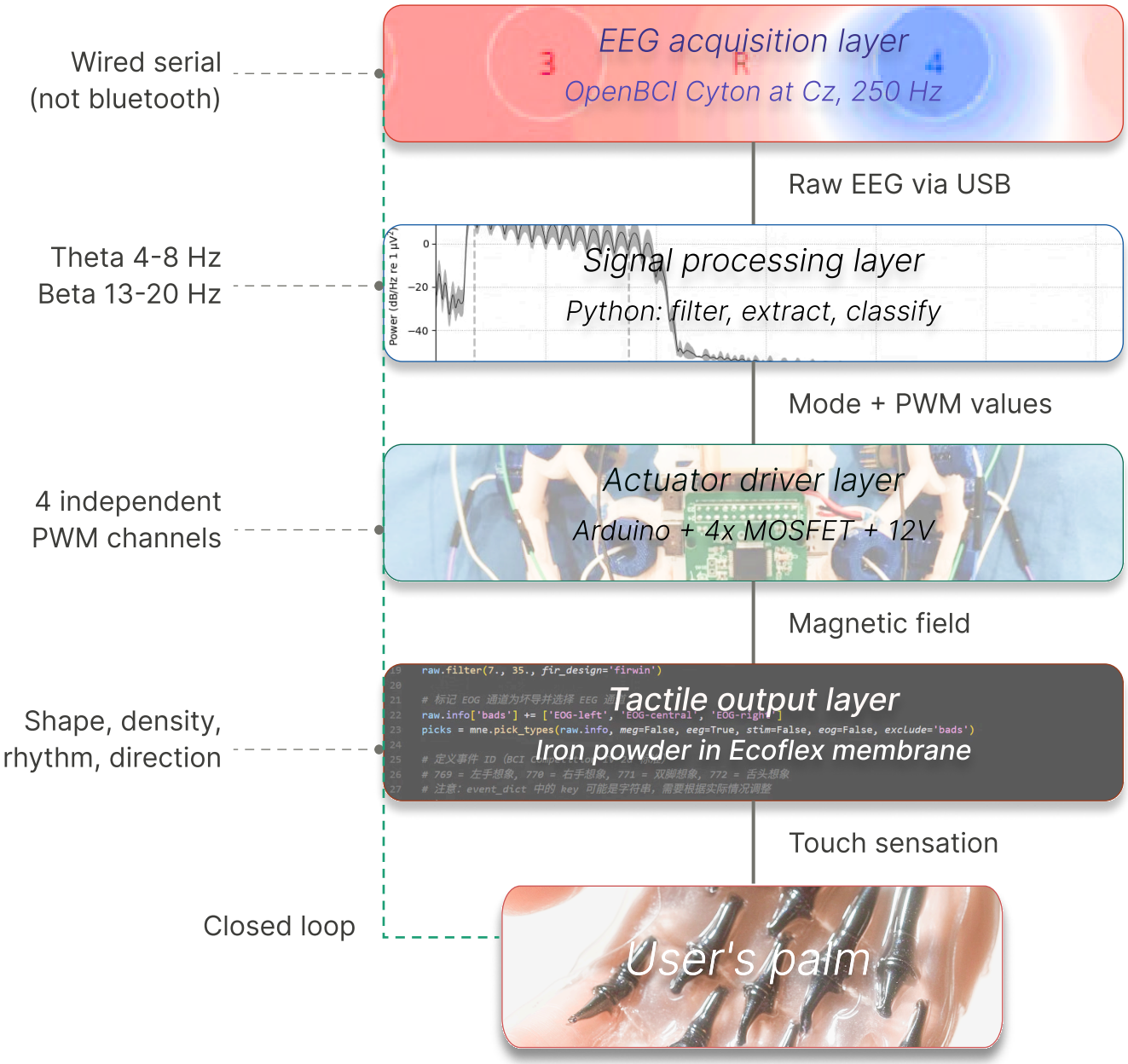
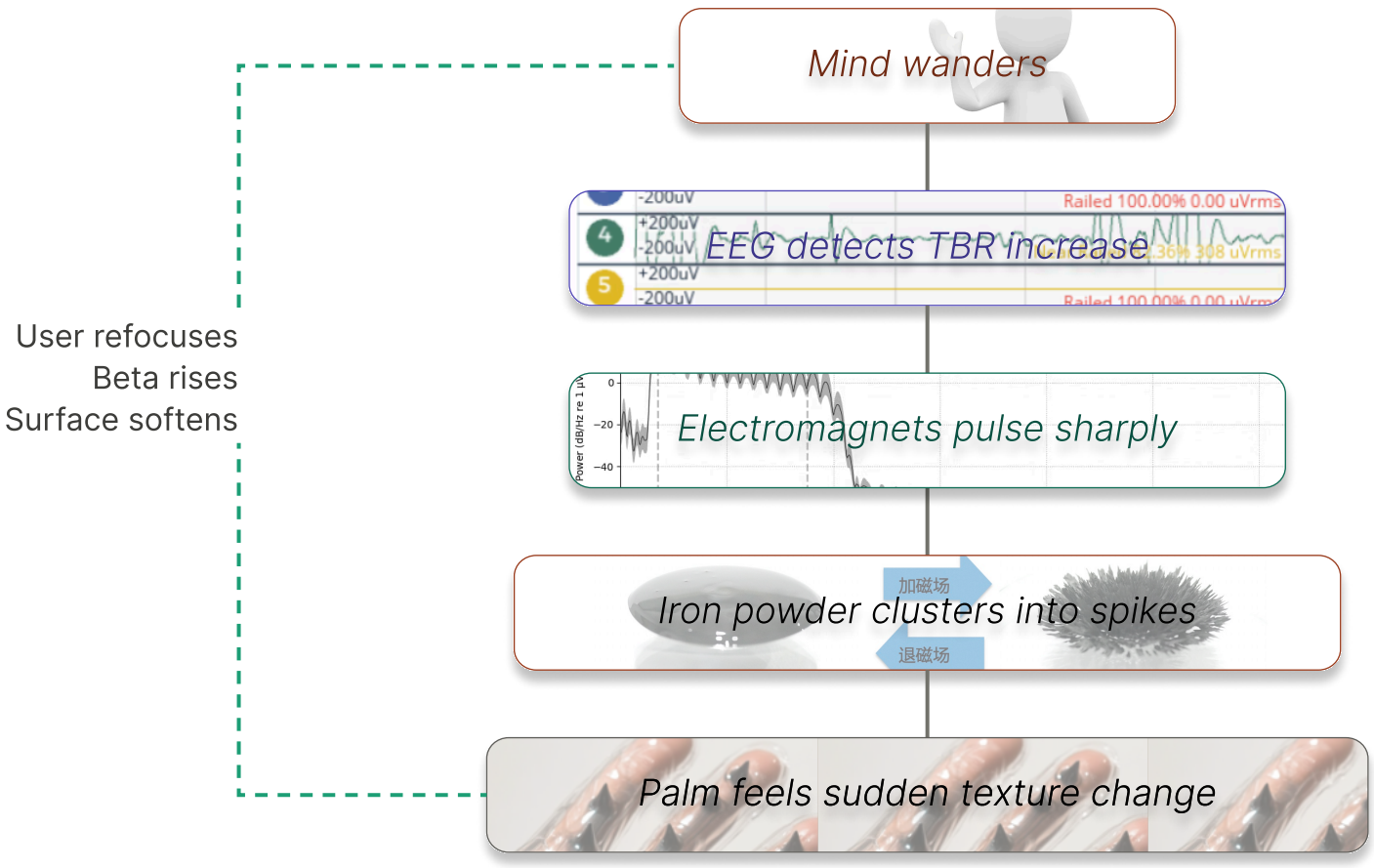
Railed 100.00% 0.00 uVrms
Not Railed 48.08% 63.6 uVrms
Railed 100.00% 0.00 uVrms
Not Railed 49.31% 22.5 uVrms
-5
-4
-3
-2
-1
0
R

System architecture



Ultra-thin PVC plastic film
Handmade silicone sheet
Film peeled off from a ready-made silicone breast pad
OR
Ferrofluid?
IRON OXIDE?
加磁场
退磁场
流体
粘弹性固体

Closed-loop neurofeedback cycle



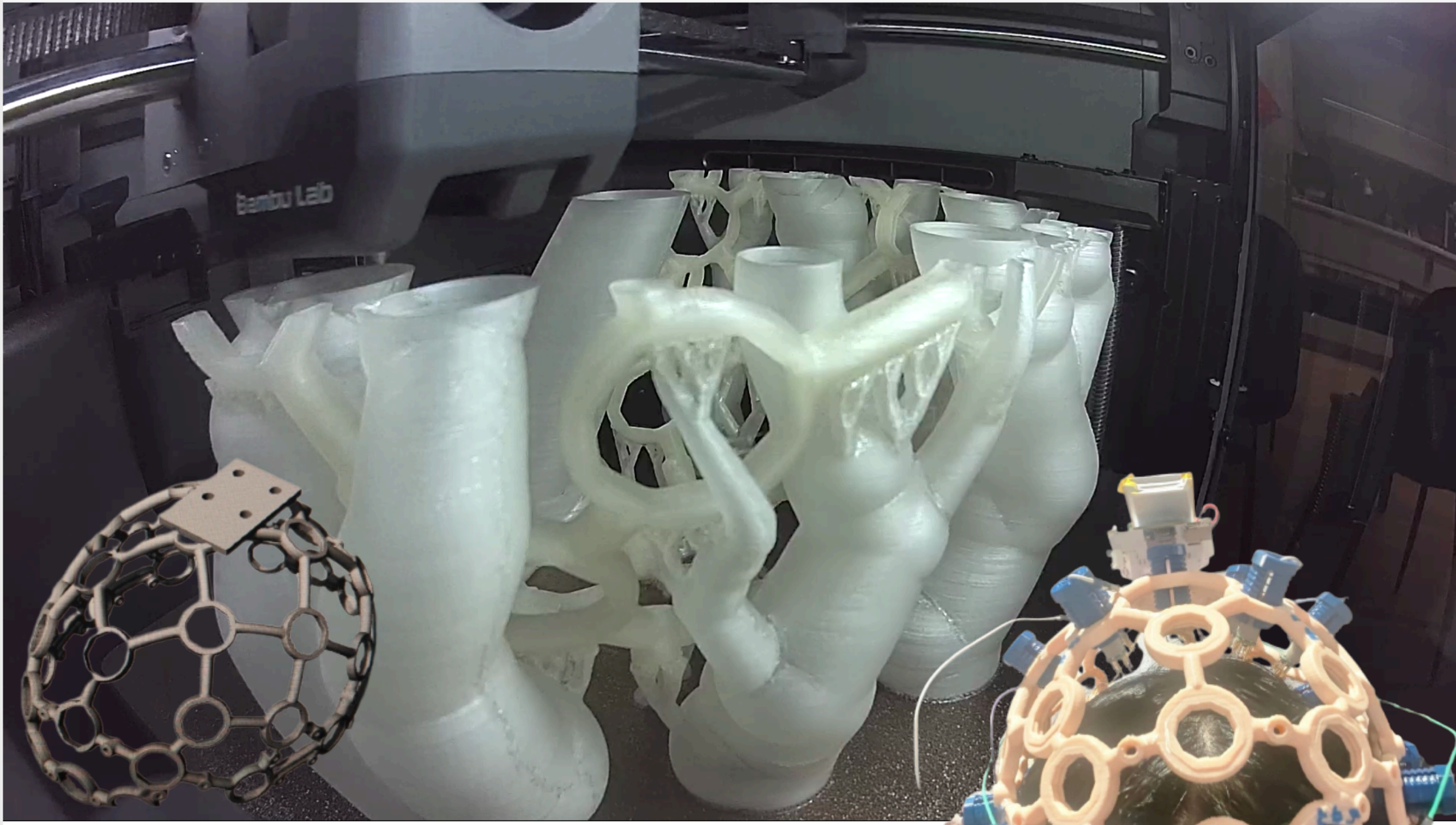
Operant conditioning:
30-40 sessions, the brain
learns to self-regulate

Effect size grows at
follow-up: SMD 0.64
to 0.80 (Van Doren 2019)

Unlike screen NF:
No visual attention
required. Hands-free
eyes. Works during
homework or reading.

Total system latency: 215 ms (within 250 ms threshold, Ossadtchi 2021)

摄像机



打印进度

1

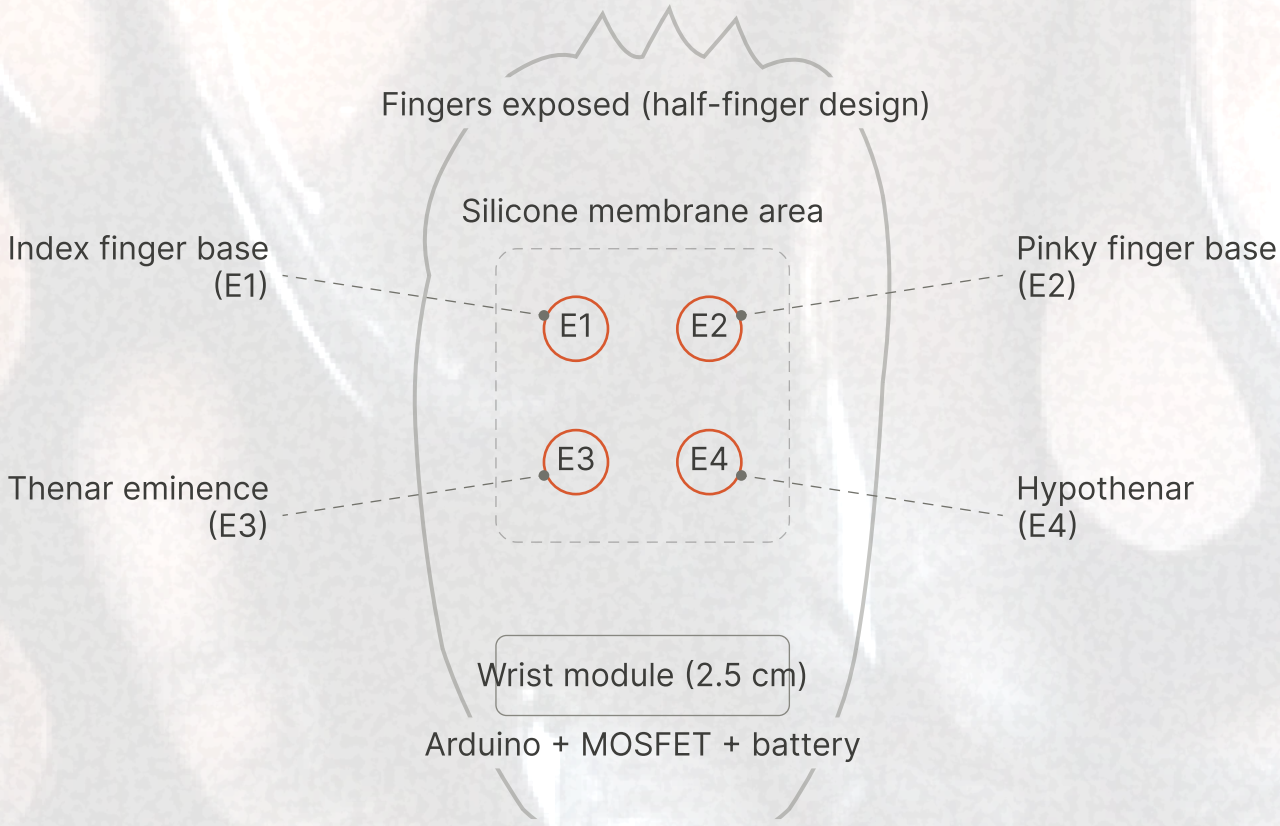
0.2mm layer, 10 walls, 99% infill

15小时10分4秒　310.28g

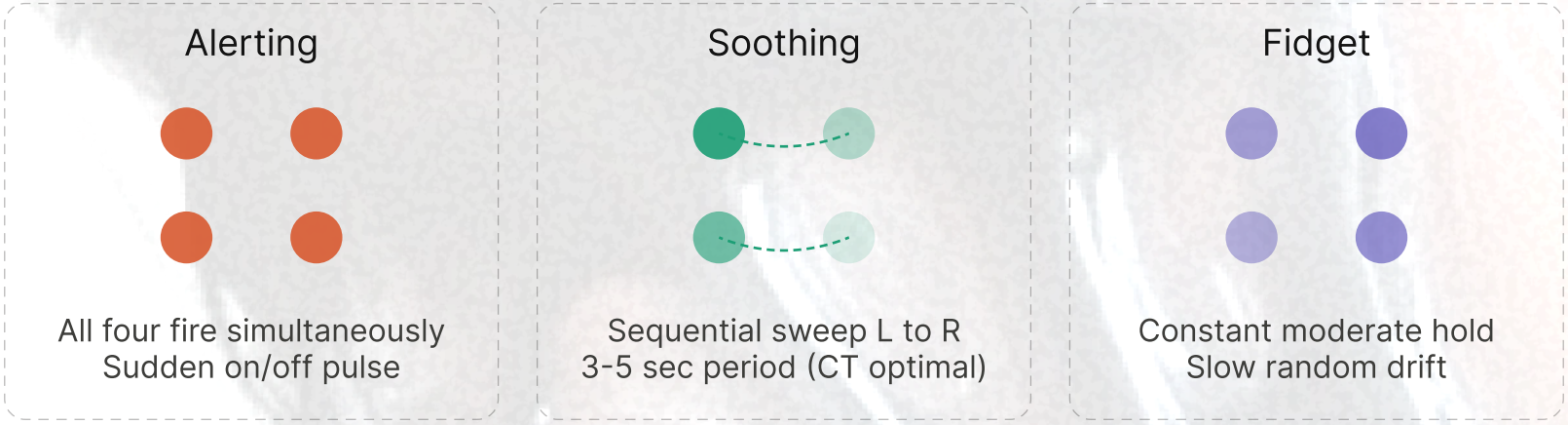
42%

层:　500/1258　-8小时41分0秒

Palm-side view: electromagnet array + iron powder membrane



Electromagnet firing patterns by mode



Circle opacity = electromagnet intensity. Same hardware, three software-defined behaviours.

train1.py load_data.py

train1.py > ...

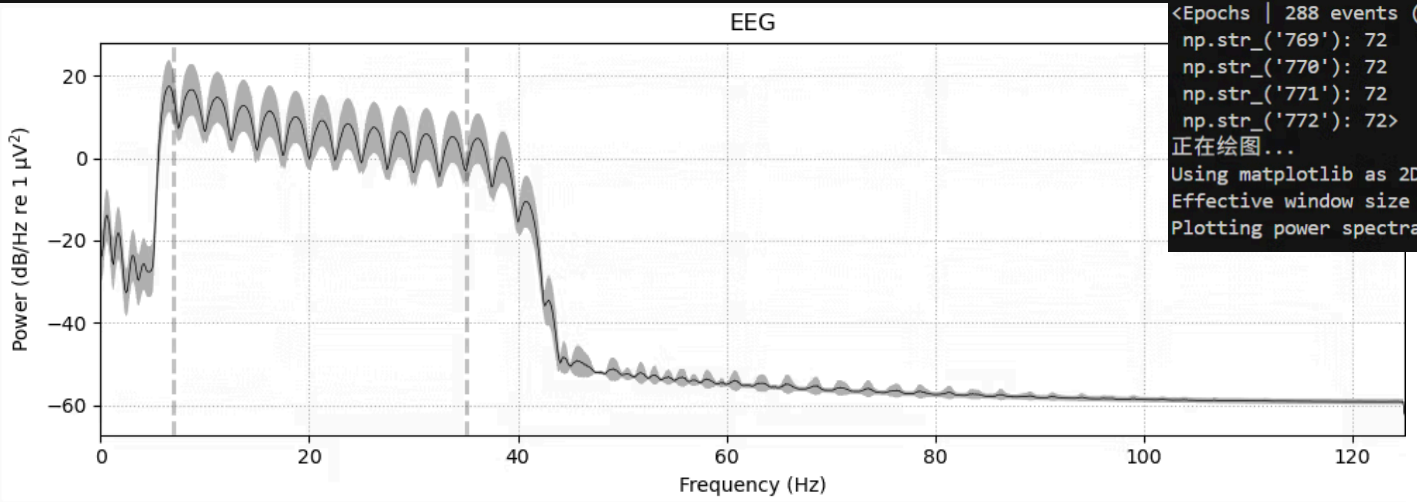
```python
# 提取事件标记
events, event_dict = mne.events_from_annotations(raw)
print("检测到的事件类型:", event_dict)

# 带通滤波 7-35 Hz（保留运动想象相关的 Mu 和 Beta 节律）
raw.filter(7., 35., fir_design='firwin')

# 标记 EOG 通道为坏导并选择 EEG 通道
raw.info['bads'] += ['EOG-left', 'EOG-central', 'EOG-right']
picks = mne.pick_types(raw.info, meg=False, eeg=True, stim=False, eog=False, exclud

# 定义事件 ID（BCI Competition IV 2a 标准）
# 769 = 左手想象, 770 = 右手想象, 771 = 双脚想象, 772 = 舌头想象
# 注意：event_dict 中的 key 可能是字符串，需要根据实际情况调整
tmin, tmax = 2, 6
event_id = {'769': event_dict.get('769', 7),
            '770': event_dict.get('770', 8),
            '771': event_dict.get('771', 9),
            '772': event_dict.get('772', 10)}

# 创建 Epochs（分段数据）
epochs = mne.Epochs(raw, events, event_id, tmin, tmax, proj=True, picks=picks, base
print(epochs)

# 绘图放在最后
raw.plot(duration=5, n_channels=22, clipping=None)
raw.compute_psd().plot(average=True)  # 使用新 API 替代 plot_psd
plt.show()
```

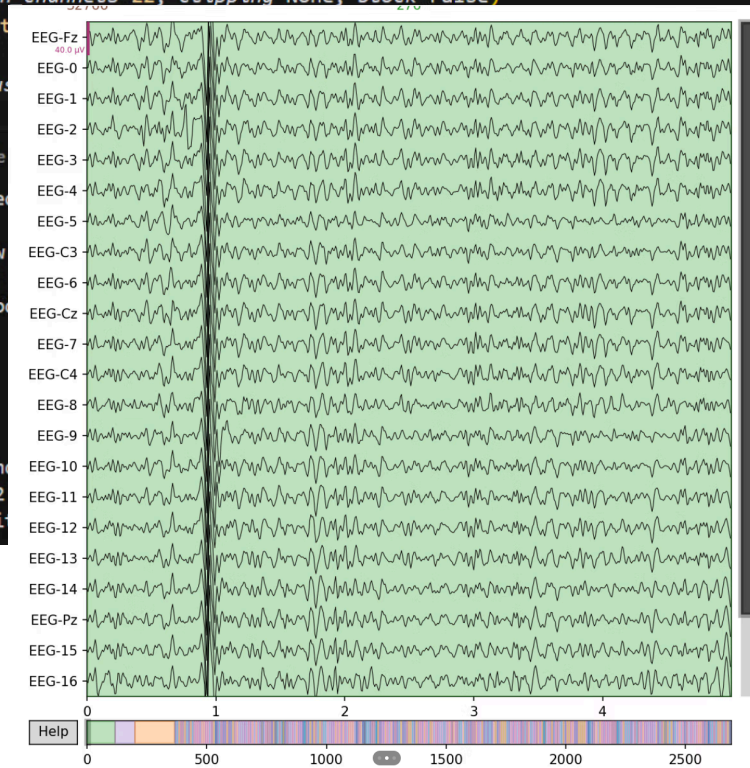
train1.py > ...

```python
# 定义事件 ID（BCI Competition IV 2a 标准）
# 769 = 左手想象, 770 = 右手想象, 771 = 双脚想象, 772 = 舌头想象
tmin, tmax = 1, 4  # 从提示后 1-4 秒（运动想象的主要时间段）

# 根据实际检测到的事件类型来设置 event_id
# 注意：event_dict 的 key 可能是字符串如 '769' 或 '770' 等
event_id = {}
for key, value in event_dict.items():
    if key in ['769', '770', '771', '772']:
        event_id[key] = value

print(f"使用的事件 ID: {event_id}", flush=True)

if len(event_id) == 0:
    print("警告：没有检测到运动想象事件 (769/770/771/772)!", flush=True)
    print("可用的事件类型:", list[Any](event_dict.keys()), flush=True)
else:
    # 创建 Epochs（分段数据）
    print("正在创建 Epochs...", flush=True)
    epochs = mne.Epochs(raw, events, event_id, tmin, tmax, proj=True,
                        picks=picks, baseline=None, preload=True)
    print(epochs, flush=True)

# 绘图放在最后
print("正在绘图...", flush=True)
raw.plot(duration=5, n_channels=22, clipping=None, block=False)
raw.compute_psd().plot
plt.show()
print("程序完成!", flu
```

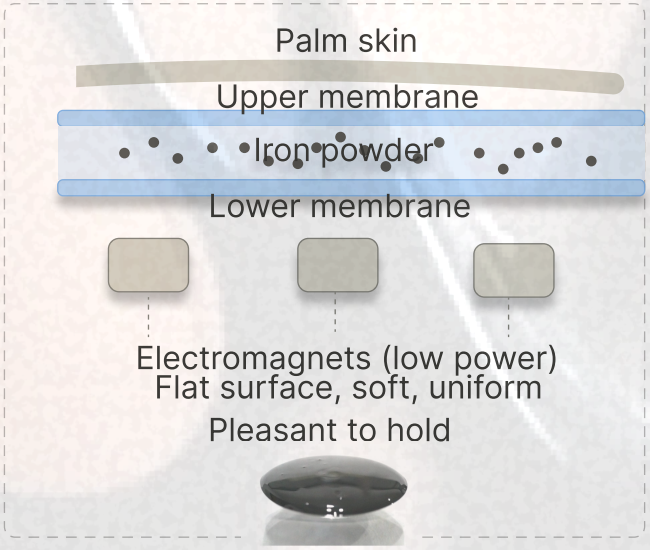
Problems Output Debug Console

```
No baseline correction applie
0 projection items activated
Using data from preloaded Raw
0 bad epochs dropped
<Epochs | 288 events (all goo
 np.str_('769'): 72
 np.str_('770'): 72
 np.str_('771'): 72
 np.str_('772'): 72>
正在绘图...
Using matplotlib as 2D backen
Effective window size : 8.192
Plotting power spectral densi
```

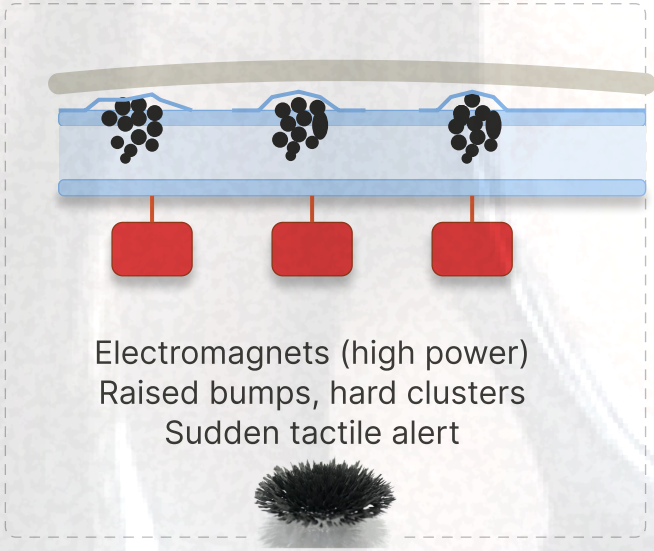




Focused state (low TBR)



Distracted state (high TBR)



Three EEG-driven tactile modes

Alerting mode
Sudden spike clusters
Theta high = distracted

Soothing mode
Slow wave + deep press
High beta + EMG = anxious

Fidget mode
Kneadable, responsive
TBR near threshold

Evidence base for each mode:
Alerting: phasic alerting via abrupt tactile change (Posner, 2008)
Soothing: CT-fiber optimal touch at 1-10 cm/s triggers oxytocin (Morrison, 2016)
Fidget: deep pressure activates parasympathetic system (Champagne, 2015)

Why this device is irreplaceable

	Fidget tools (Tangle, putty)	Screen-based neurofeedback	Vibrotactile feedback	This device
Feedback channel	Tactile (passive)	Visual (screen)	Tactile (vibration)	Tactile (shape-change)
Closed-loop?	No	Yes	Yes	Yes
Requires screen?	No	Yes	No	No
Tactile dimensions	1D (squeeze)	None	1D (on/off/intensity)	4D (position, density, rhythm, direction)
Habituation risk	High (effect gone by 2nd use)	Low	Medium	Low (pattern never repeats
Lasting training?	No	Yes (sustained at follow-up)	Unverified	Yes (NF mechanism)
Adaptive modes	None	1 (visual reward)	1 (vibration intensity)	3 (alert, soothe, fidget)

Vibration motors have 1 dimension: buzz intensity. Iron powder in membrane has 4: where the particles gather, how densely they cluster, how fast they pulse, and which direction they flow across the palm.